CSCI1409 Deep Learning · Week 4

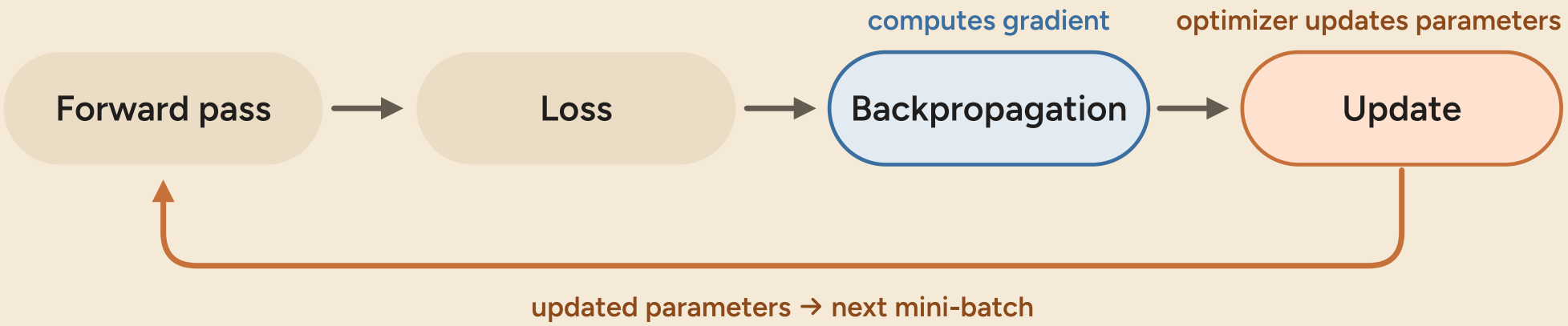
Optimization Methods for Deep Learning

12:00–13:00 · Lecture

13:00–14:00 · Quiz

Aidyn Abirov · Fall 2026–2027

One training step



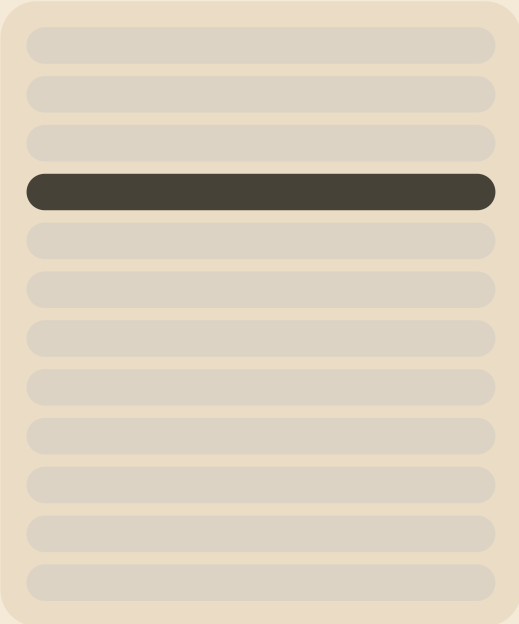
$$g_t = \nabla_\theta L_{B_t}(\theta_t)$$

Gradient = direction of rising loss; **update** moves against it.

Which data goes into a step?



Full batch: all N examples



Single example: one example



Mini-batch: subset B_t of the data

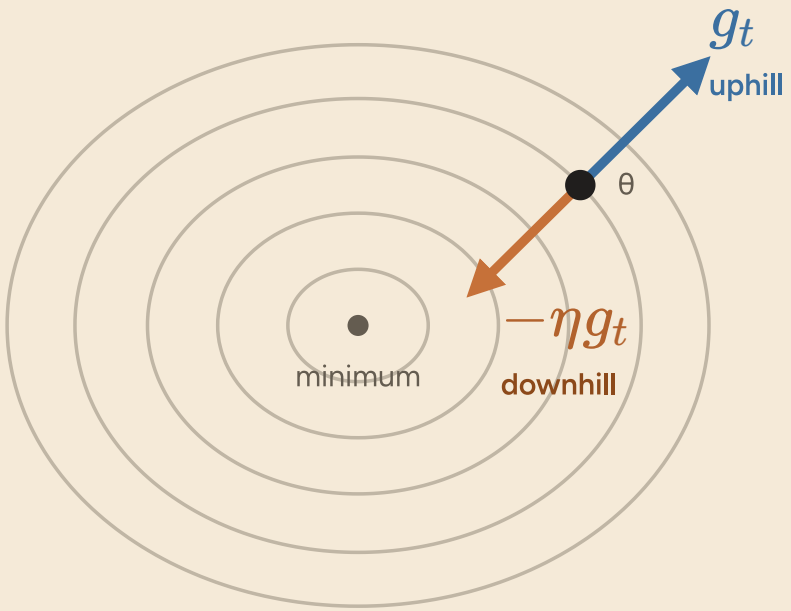
Mini-batch training is often called SGD in practice.

Stochastic gradient descent

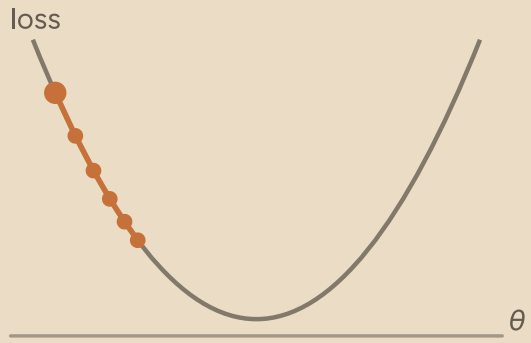
$$g_t = \frac{1}{|B_t|} \sum_{i \in B_t} \nabla_\theta \ell_i(\theta_t)$$

$$\theta_{t+1} = \theta_t - \eta\, g_t$$

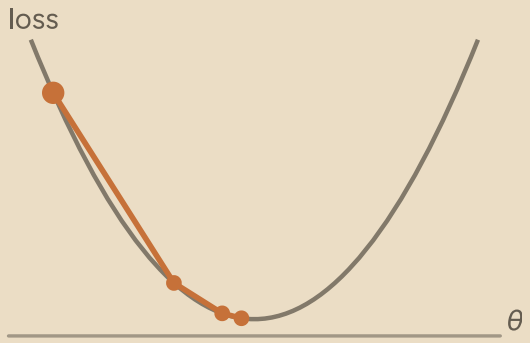
θ: weights and biases · η: learning rate



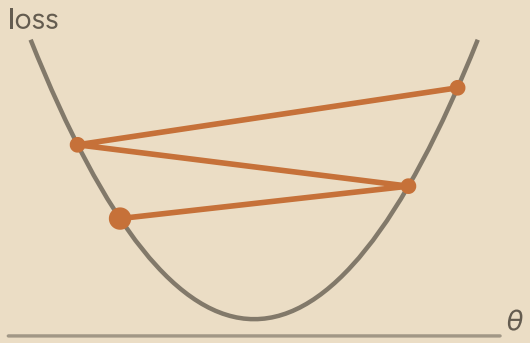
How large should the step be?



Small η → slow progress



Productive steps



Excessive η → overshoot or divergence

Learning rate changes the size of $-\eta\, g_t$.

One exact gradient step

$$L(\theta) = (\theta - 3)^2, \quad \theta_0 = 0, \quad g_0 = -6, \quad L(0) = 9$$

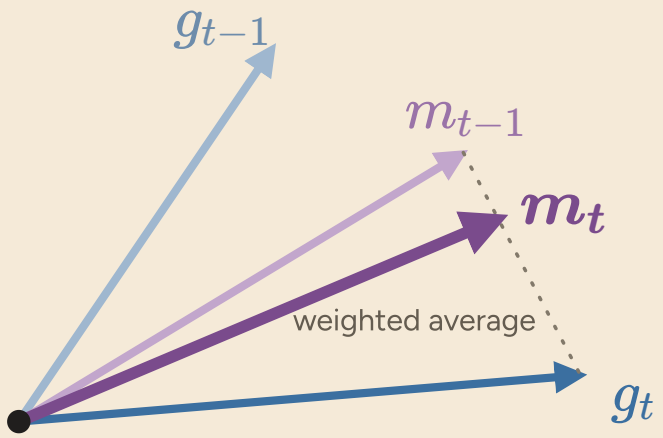
η	θ_1	$L(\theta_1)$
0.1	0.6	5.76
0.5	3.0	0
1.1	6.6	12.96

Momentum keeps a memory

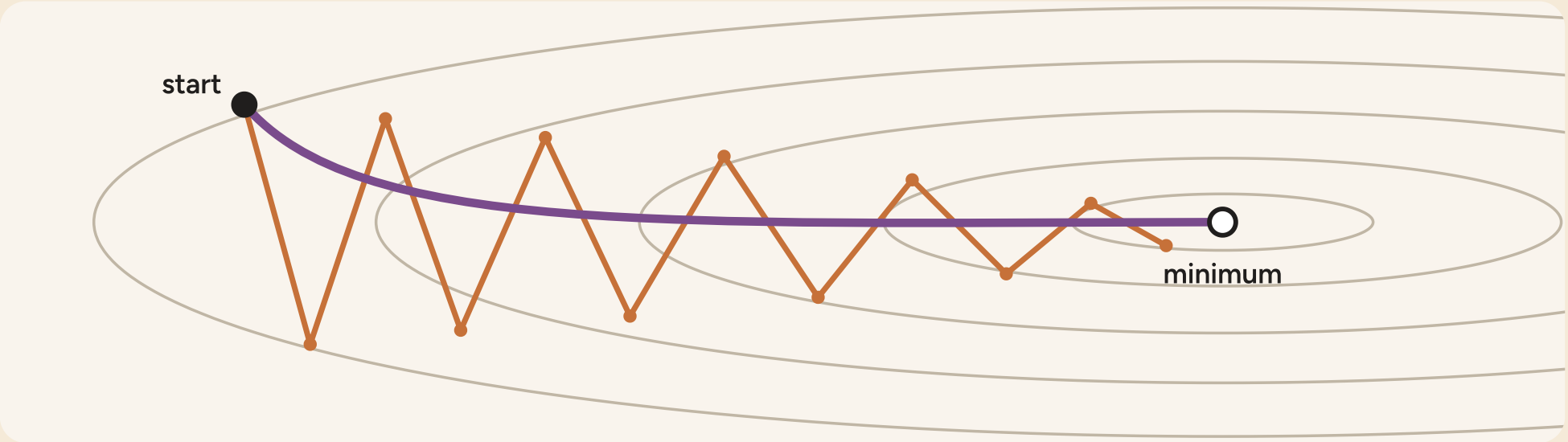
$$m_t = \beta\, m_{t-1} + (1 - \beta)\, g_t$$

$$\theta_{t+1} = \theta_t - \eta\, m_t$$

β: how much of the previous direction remains



Why momentum can help



Plain SGD: back-and-forth steps across a narrow valley

Momentum: smoother direction along the valley

It helps in this setting; results depend on the problem.

Adam stores two moving averages

gradient average m

squared-gradient average v

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\, g_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)\, g_t^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

Bias correction compensates for starting m and v at zero.

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

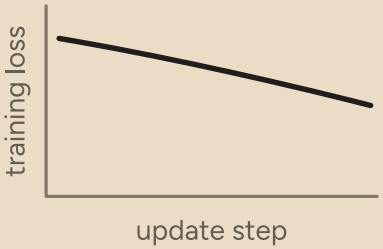
Adam's first step, checked

1. $g_1 = 2, \quad \beta_1 = 0.9, \quad \beta_2 = 0.999$
2. $m_1 = 0.2, \quad v_1 = 0.004$
3. $\hat{m}_1 = 2, \quad \hat{v}_1 = 4$
4. $\Delta\theta = -\eta\, 2/(2+\epsilon) \approx -\eta$

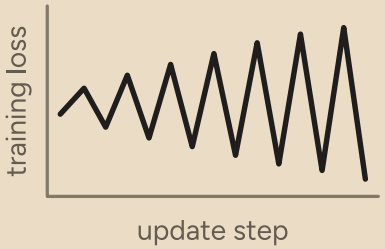
Compare the update rules

Method	Past gradients	Per-parameter scale
SGD	○ No	○ No
Momentum	● **Yes**	○ No
Adam	● **Yes**	● **Yes**, from squared gradients

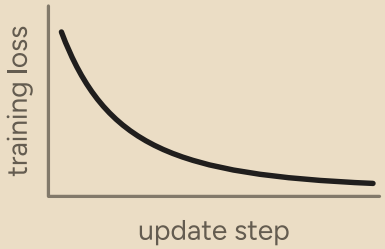
Read your training curves



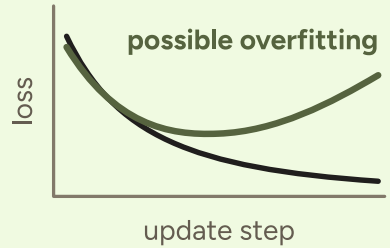
Slowly falling loss → try a larger learning rate cautiously



Oscillating or growing loss → try a smaller learning rate



Useful decline



Monitor **validation loss** as well as training loss.

Takeaways

- Backprop **computes gradients**; the optimizer **updates parameters**.
- Learning rate controls the step; **momentum remembers gradients**.
- Adam also tracks **squared gradients** and corrects startup bias.

13:00–14:00 · Quiz